

Mathematical Notes on Expectation Maximization

Based on Week 3 Lecture Materials

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1 Latent Variable Models (continued)

1.1 Log-Likelihoods in Exponential Families

Definition 1.1 (Exponential Family Model). An exponential family model has the form

$$p(\mathbf{x}|\boldsymbol{\theta}) = f(\mathbf{x})e^{\boldsymbol{\theta}^T T(\mathbf{x})}/Z(\boldsymbol{\theta}) \quad (1)$$

where $T(\mathbf{x})$ is the sufficient statistic and $Z(\boldsymbol{\theta})$ is the partition function.

Proposition 1.2 (Exponential Family Log-Likelihood). *For an exponential family model (Definition 1.1) with iid observations $\mathbf{x}_1, \dots, \mathbf{x}_N$, the log-likelihood is*

$$\ell(\boldsymbol{\theta}) = \boldsymbol{\theta}^T \sum_n T(\mathbf{x}_n) - N \log Z(\boldsymbol{\theta}) + \text{constants} \quad (2)$$

This is a concave function of $\boldsymbol{\theta}$.

Remark 1.3 (Tractability of Exponential Families). For exponential family models without latent variables:

- The maximum may have a closed-form solution
- If not, numerical optimization is generally straightforward

1.2 Latent Variable Models

Definition 1.4 (Latent Variable Model). A latent variable model has the form

$$p(\mathbf{x}|\boldsymbol{\theta}_x, \boldsymbol{\theta}_z) = \int d\mathbf{z} f_x(\mathbf{x}) \frac{e^{\phi(\boldsymbol{\theta}_x, \mathbf{z})^T T_x(\mathbf{x})}}{Z_x(\phi(\boldsymbol{\theta}_x, \mathbf{z}))} f_z(\mathbf{z}) \frac{e^{\boldsymbol{\theta}_z^T T_z(\mathbf{z})}}{Z_z(\boldsymbol{\theta}_z)} \quad (3)$$

where $\mathbf{z}$ is the latent variable.

Proposition 1.5 (Latent Variable Log-Likelihood). *The log-likelihood for a latent variable model (Definition 1.4) is*

$$\ell(\boldsymbol{\theta}_x, \boldsymbol{\theta}_z) = \sum_n \log \int d\mathbf{z} f_x(\mathbf{x}) \frac{e^{\phi(\boldsymbol{\theta}_x, \mathbf{z})^T T_x(\mathbf{x})}}{Z_x(\phi(\boldsymbol{\theta}_x, \mathbf{z}))} f_z(\mathbf{z}) \frac{e^{\boldsymbol{\theta}_z^T T_z(\mathbf{z})}}{Z_z(\boldsymbol{\theta}_z)} \quad (4)$$

Remark 1.6 (Intractability of Latent Variable Models). For latent variable models (Definition 1.4):

- Usually no closed-form optimum
- Often multiple local maxima
- Direct numerical optimization may be possible but infrequently easy

This stands in contrast to the exponential family case without latent variables (Remark 1.3).

1.3 Mixture of Gaussians

Example 1.7 (Mixture of Gaussians). Data: $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

Latent process:

$$s_i \stackrel{\text{iid}}{\sim} \text{Disc}[\boldsymbol{\pi}]$$

Component distributions:

$$\mathbf{x}_i | (s_i = m) \sim \mathcal{P}_m[\boldsymbol{\theta}_m] = \mathcal{N}(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$$

Marginal distribution:

$$P(\mathbf{x}_i) = \sum_{m=1}^k \pi_m P_m(\mathbf{x}; \boldsymbol{\theta}_m) \quad (5)$$

Log-likelihood:

$$\ell(\{\boldsymbol{\mu}_m\}, \{\boldsymbol{\Sigma}_m\}, \boldsymbol{\pi}) = \sum_{i=1}^n \log \sum_{m=1}^k \frac{\pi_m}{\sqrt{|2\pi\boldsymbol{\Sigma}_m|}} e^{-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_m)^T \boldsymbol{\Sigma}_m^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_m)} \quad (6)$$

This is an instance of a latent variable model (Definition 1.4), where the latent variable s_i selects the mixture component.

1.4 Motivation for EM

Remark 1.8 (Joint-Data Likelihood). For many models, maximization might be straightforward if $\mathbf{z}$ were not latent, and we could just maximize the joint-data likelihood:

$$\ell(\boldsymbol{\theta}_x, \boldsymbol{\theta}_z) = \sum_n \phi(\boldsymbol{\theta}_x, \mathbf{z}_n)^T T_x(\mathbf{x}_n) + \boldsymbol{\theta}_z^T \sum_n T_z(\mathbf{z}_n) - \sum_n \log Z_x(\phi(\boldsymbol{\theta}_x, \mathbf{z}_n)) - N \log Z_z(\boldsymbol{\theta}_z) \quad (7)$$

Remark 1.9 (Known Parameters). Conversely, if we knew $\boldsymbol{\theta}$, we might easily compute (the posterior over) the values of $\mathbf{z}$.

Remark 1.10 (Key Idea). Update $\boldsymbol{\theta}$ and (the distribution on) $\mathbf{z}$ in alternation, to reach a self-consistent answer. This is the Expectation Maximization (EM) algorithm.

2 The Expectation Maximization Algorithm

Definition 2.1 (EM Algorithm). The EM algorithm (Dempster, Laird & Rubin, 1977) finds a (local) maximum of a latent variable model likelihood (Proposition 1.5). Starting from arbitrary values of the parameters, iterate two steps:

E step: Fill in values of latent variables according to posterior given data.

M step: Maximize likelihood as if latent variables were not hidden.

Remark 2.2 (Properties of EM). • Decomposes difficult problems into series of tractable steps

- An alternative to gradient-based iterative methods
- No learning rate required

- In ML, the E step is called *inference*, and the M step *learning*
- In statistics, these are often *imputation* and *inference* or *estimation*
- Not essential for simple models (like MoGs/FA), though often more efficient
- Crucial for learning in complex settings
- Provides a framework for principled approximations

2.1 Jensen's Inequality

Lemma 2.3 (Jensen's Inequality). *For $\alpha_i \geq 0$ with $\sum \alpha_i = 1$ and $\{x_i > 0\}$:*

$$\log \left(\sum_i \alpha_i x_i \right) \geq \sum_i \alpha_i \log(x_i) \quad (8)$$

More generally, for a probability measure α and concave function f :

$$f(\mathbb{E}_\alpha[x]) \geq \mathbb{E}_\alpha[f(x)] \quad (9)$$

Equality holds if and only if $f(x)$ is almost surely constant or linear on the (convex) support of α .

Remark 2.4 (Jensen and EM). Jensen's inequality (Lemma 2.3) shows that EM iteratively refines a lower bound on the log-likelihood.

2.2 The Free Energy Lower Bound

Definition 2.5 (Free Energy). Let $\mathcal{X} = \{\mathbf{x}_i\}$ be observed data, $\mathcal{Z} = \{\mathbf{z}_i\}$ be latent variables, and $\boldsymbol{\theta} = \{\boldsymbol{\theta}_x, \boldsymbol{\theta}_z\}$ be parameters.

The log-likelihood is:

$$\ell(\boldsymbol{\theta}) = \log P(\mathcal{X}|\boldsymbol{\theta}) = \log \int d\mathcal{Z} P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) \quad (10)$$

Proposition 2.6 (Free Energy Lower Bound). *For any distribution $q(\mathcal{Z})$ over the latent variables:*

$$\ell(\boldsymbol{\theta}) = \log \int d\mathcal{Z} q(\mathcal{Z}) \frac{P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta})}{q(\mathcal{Z})} \geq \int d\mathcal{Z} q(\mathcal{Z}) \log \frac{P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta})}{q(\mathcal{Z})} \equiv \mathcal{F}(q, \boldsymbol{\theta}) \quad (11)$$

by Jensen's inequality (Lemma 2.3).

Proposition 2.7 (Free Energy Decomposition). *The free energy (Proposition 2.6) can be written as:*

$$\mathcal{F}(q, \boldsymbol{\theta}) = \langle \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) \rangle_{q(\mathcal{Z})} + H_q \quad (12)$$

where $H_q = - \int d\mathcal{Z} q(\mathcal{Z}) \log q(\mathcal{Z})$ is the entropy of $q(\mathcal{Z})$.

Proof.

$$\begin{aligned} \int d\mathcal{Z} q(\mathcal{Z}) \log \frac{P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta})}{q(\mathcal{Z})} &= \int d\mathcal{Z} q(\mathcal{Z}) \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) - \int d\mathcal{Z} q(\mathcal{Z}) \log q(\mathcal{Z}) \\ &= \int d\mathcal{Z} q(\mathcal{Z}) \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) + H_q \end{aligned}$$

□

2.3 The E and M Steps

Definition 2.8 (EM Steps via Free Energy). The free-energy lower bound on $\ell(\boldsymbol{\theta})$ (Proposition 2.6) is a function of $\boldsymbol{\theta}$ and a distribution q :

$$\mathcal{F}(q, \boldsymbol{\theta}) = \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{q(\mathcal{Z})} + H_q$$

The EM steps are:

E step: Optimize $\mathcal{F}(q, \boldsymbol{\theta})$ with respect to the distribution over hidden variables, holding parameters fixed:

$$q^{(k)}(\mathcal{Z}) := \arg \max_{q(\mathcal{Z})} \mathcal{F}(q(\mathcal{Z}), \boldsymbol{\theta}^{(k-1)}) \quad (13)$$

M step: Maximize $\mathcal{F}(q, \boldsymbol{\theta})$ with respect to parameters, holding hidden distribution fixed:

$$\boldsymbol{\theta}^{(k)} := \arg \max_{\boldsymbol{\theta}} \mathcal{F}(q^{(k)}(\mathcal{Z}), \boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{q^{(k)}(\mathcal{Z})} \quad (14)$$

The second equality comes from the fact that $H_{q^{(k)}}(\mathcal{Z})$ does not depend directly on $\boldsymbol{\theta}$.

2.4 The E Step in Detail

Proposition 2.9 (Free Energy and KL Divergence). *The free energy (Proposition 2.6) can be rewritten as:*

$$\begin{aligned} \mathcal{F}(q, \boldsymbol{\theta}) &= \int q(\mathcal{Z}) \log \frac{P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta})}{q(\mathcal{Z})} d\mathcal{Z} \\ &= \int q(\mathcal{Z}) \log \frac{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}) P(\mathcal{X} | \boldsymbol{\theta})}{q(\mathcal{Z})} d\mathcal{Z} \\ &= \int q(\mathcal{Z}) \log P(\mathcal{X} | \boldsymbol{\theta}) d\mathcal{Z} + \int q(\mathcal{Z}) \log \frac{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta})}{q(\mathcal{Z})} d\mathcal{Z} \\ &= \ell(\boldsymbol{\theta}) - KL[q(\mathcal{Z}) \| P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta})] \end{aligned}$$

Theorem 2.10 (Optimal E Step). *For fixed $\boldsymbol{\theta}$, $\mathcal{F}$ is bounded above by ℓ , and achieves that bound when*

$$KL[q(\mathcal{Z}) \| P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta})] = 0$$

Since $KL[q \| p] = 0$ if and only if $q = p$ (Lemma 9.1), the E step sets

$$q^{(k)}(\mathcal{Z}) = P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^{(k-1)}) \quad (15)$$

and after an E step, the free energy equals the likelihood.

2.5 Monotonicity of EM

Theorem 2.11 (EM Never Decreases Likelihood). *The E and M steps (Definition 2.8) together never decrease the log-likelihood:*

$$\ell(\boldsymbol{\theta}^{(k-1)}) \stackrel{E \text{ step}}{=} \mathcal{F}(q^{(k)}, \boldsymbol{\theta}^{(k-1)}) \stackrel{M \text{ step}}{\leq} \mathcal{F}(q^{(k)}, \boldsymbol{\theta}^{(k)}) \stackrel{Jensen}{\leq} \ell(\boldsymbol{\theta}^{(k)}) \quad (16)$$

Proof. • The E step brings the free energy to the likelihood (Theorem 2.10), by setting $q = P(\mathcal{Z}|\mathcal{X}, \boldsymbol{\theta})$

- The M-step maximizes the free energy with respect to $\boldsymbol{\theta}$ ((14))
- $\mathcal{F} \leq \ell$ by Jensen's inequality (Lemma 2.3), or equivalently, from the non-negativity of KL divergence (Lemma 9.1)

If the M-step is executed so that $\boldsymbol{\theta}^{(k)} \neq \boldsymbol{\theta}^{(k-1)}$ if and only if $\mathcal{F}$ increases, then the overall EM iteration will step to a new value of $\boldsymbol{\theta}$ if and only if the likelihood increases. $\square$

2.6 Convergence Properties

Theorem 2.12 (Fixed Points of EM). *Fixed points of EM (generally) correspond to maxima of the likelihood.*

Proof sketch. See appendix for detailed proof that:

1. Fixed points of EM are stationary points of $\ell(\boldsymbol{\theta})$ (Theorem 9.2)
2. Maxima in $\mathcal{F}$ correspond to maxima in ℓ (Theorem 9.3)

$\square$

3 EM Summary

Theorem 3.1 (EM Algorithm Summary). *EM is an iterative algorithm that finds (local) maxima of the likelihood of a latent variable model (Definition 1.4):*

$$\ell(\boldsymbol{\theta}) = \log P(\mathcal{X}|\boldsymbol{\theta}) = \log \int d\mathcal{Z} P(\mathcal{X}|\mathcal{Z}, \boldsymbol{\theta}) P(\mathcal{Z}|\boldsymbol{\theta}) \quad (17)$$

It increases a variational lower bound on the likelihood by coordinate ascent (Proposition 2.9):

$$\mathcal{F}(q, \boldsymbol{\theta}) = \langle \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) \rangle_{q(\mathcal{Z})} + H_q = \ell(\boldsymbol{\theta}) - KL[q(\mathcal{Z}) \| P(\mathcal{Z}|\mathcal{X})] \leq \ell(\boldsymbol{\theta}) \quad (18)$$

E step:

$$q^{(k)}(\mathcal{Z}) := \arg \max_{q(\mathcal{Z})} \mathcal{F}(q(\mathcal{Z}), \boldsymbol{\theta}^{(k-1)}) = P(\mathcal{Z}|\mathcal{X}, \boldsymbol{\theta}^{(k-1)})$$

M step:

$$\boldsymbol{\theta}^{(k)} := \arg \max_{\boldsymbol{\theta}} \mathcal{F}(q^{(k)}(\mathcal{Z}), \boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \langle \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) \rangle_{q^{(k)}(\mathcal{Z})}$$

After the E-step, $\mathcal{F}(q, \boldsymbol{\theta}) = \ell(\boldsymbol{\theta})$ (Theorem 2.10), so the maximum of the free-energy is a maximum of the likelihood (Theorem 9.3).

4 Variants of EM

4.1 Partial M Steps

Definition 4.1 (Generalized EM). The proof of monotonicity (Theorem 2.11) holds even if we just *increase* $\mathcal{F}$ with respect to θ rather than maximize it. This is called the generalized EM, or GEM, algorithm.

Proposition 4.2 (Gradient After E Step). *Immediately after an E step (as in Definition 2.8):*

$$\left. \frac{\partial}{\partial \theta} \right|_{\theta^{(k-1)}} \langle \log P(\mathcal{X}, \mathcal{Z} | \theta) \rangle_{q^{(k)}(\mathcal{Z}) [= P(\mathcal{Z} | \mathcal{X}, \theta^{(k-1)})]} = \left. \frac{\partial}{\partial \theta} \right|_{\theta^{(k-1)}} \log P(\mathcal{X} | \theta) \quad (19)$$

Remark 4.3 (Expectation Conjugate Gradient). The E-step (inference) can be used to construct other gradient-based optimization schemes (e.g., “Expectation Conjugate Gradient”, Salakhutdinov et al. ICML 2003).

4.2 Partial E Steps

Remark 4.4 (Partial E Steps). We can also just *increase* $\mathcal{F}$ with respect to some of the q ’s. For example:

- Sparse or online versions of the EM algorithm would compute the posterior for a subset of the data points or as the data arrives
- One might also update the posterior over a subset of the hidden variables, while holding others fixed

5 Application: Mixture of Gaussians

5.1 The Gaussian Mixture Model E-Step

Example 5.1 (Univariate Gaussian Mixture E-Step). In a univariate Gaussian mixture model (cf. Example 1.7), the density of a data point x is:

$$p(x | \theta) = \sum_{m=1}^k p(s = m | \theta) p(x | s = m, \theta) \propto \sum_{m=1}^k \frac{\pi_m}{\sigma_m} \exp \left\{ -\frac{1}{2\sigma_m^2} (x - \mu_m)^2 \right\} \quad (20)$$

where θ is the collection of parameters: means μ_m , variances σ_m^2 , and mixing proportions $\pi_m = p(s = m | \theta)$.

The hidden variable s_i indicates which component generated observation x_i .

The E-step (Definition 2.8) computes the posterior for s_i given the current parameters:

$$q(s_i) = p(s_i | x_i, \theta) \propto p(x_i | s_i, \theta) p(s_i | \theta)$$

Define the *responsibilities*:

$$r_{im} \equiv q(s_i = m) \propto \frac{\pi_m}{\sigma_m} \exp \left\{ -\frac{1}{2\sigma_m^2} (x_i - \mu_m)^2 \right\} = \langle \delta_{s_i=m} \rangle_q \quad (21)$$

with normalization $\sum_m r_{im} = 1$.

5.2 The Gaussian Mixture Model M-Step

Example 5.2 (Univariate Gaussian Mixture M-Step). In the M-step (Definition 2.8) we optimize the sum (since s is discrete):

$$\begin{aligned} E &= \langle \log p(\mathbf{x}, \mathbf{s} | \boldsymbol{\theta}) \rangle_{q(\mathbf{s})} = \sum_{\mathbf{s}} q(\mathbf{s}) \log[p(\mathbf{s} | \boldsymbol{\theta}) p(\mathbf{x} | \mathbf{s}, \boldsymbol{\theta})] \\ &= \sum_{i,m} r_{im} \left[\log \pi_m - \log \sigma_m - \frac{1}{2\sigma_m^2} (x_i - \mu_m)^2 \right] \end{aligned}$$

The optimum is found by setting the partial derivatives of E to zero:

$$\frac{\partial}{\partial \mu_m} E = \sum_i r_{im} \frac{(x_i - \mu_m)}{2\sigma_m^2} = 0 \quad \Rightarrow \quad \mu_m = \frac{\sum_i r_{im} x_i}{\sum_i r_{im}} \quad (22)$$

$$\frac{\partial}{\partial \sigma_m} E = \sum_i r_{im} \left[-\frac{1}{\sigma_m} + \frac{(x_i - \mu_m)^2}{\sigma_m^3} \right] = 0 \quad \Rightarrow \quad \sigma_m^2 = \frac{\sum_i r_{im} (x_i - \mu_m)^2}{\sum_i r_{im}} \quad (23)$$

$$\frac{\partial}{\partial \pi_m} E = \sum_i r_{im} \frac{1}{\pi_m}, \quad \frac{\partial E}{\partial \pi_m} + \lambda = 0 \quad \Rightarrow \quad \pi_m = \frac{1}{n} \sum_i r_{im} \quad (24)$$

where λ is a Lagrange multiplier ensuring that the mixing proportions sum to unity.

Remark 5.3 (Multivariate Case). For multivariate Gaussians (cf. Example 1.7), the EM updates are:

E-step: Evaluate responsibilities (as in (21))

$$r_{im} = \frac{P_m(\mathbf{x}_i) \pi_m}{\sum_{m'} P_{m'}(\mathbf{x}_i) \pi_{m'}} \quad (25)$$

M-step: Update parameters (cf. (22)–(24))

$$\boldsymbol{\mu}_m \leftarrow \frac{\sum_i r_{im} \mathbf{x}_i}{\sum_i r_{im}} \quad (26)$$

$$\boldsymbol{\Sigma}_m \leftarrow \frac{\sum_i r_{im} (\mathbf{x}_i - \boldsymbol{\mu}_m)(\mathbf{x}_i - \boldsymbol{\mu}_m)^T}{\sum_i r_{im}} \quad (27)$$

$$\pi_m \leftarrow \frac{\sum_i r_{im}}{N} \quad (28)$$

6 EM for Exponential Families

Theorem 6.1 (EM for Exponential Family Models). *EM is often applied to models whose joint over $\xi = (\mathbf{z}, \mathbf{x})$ has exponential-family form (Definition 1.1):*

$$p(\xi | \boldsymbol{\theta}) = f(\xi) \exp\{\boldsymbol{\theta}^T T(\xi)\} / Z(\boldsymbol{\theta}) \quad (29)$$

but whose marginal $p(\mathbf{x}) \notin \text{ExpFam}$.

The free energy dependence on $\boldsymbol{\theta}$ is given by (using (12)):

$$\begin{aligned} \mathcal{F}(q, \boldsymbol{\theta}) &= \int q(\mathbf{z}) \log p(\mathbf{z}, \mathbf{x} | \boldsymbol{\theta}) d\mathbf{z} + H_q \\ &= \int q(\mathbf{z}) [\boldsymbol{\theta}^T T(\mathbf{z}, \mathbf{x}) - \log Z(\boldsymbol{\theta})] d\mathbf{z} + \text{const wrt } \boldsymbol{\theta} \\ &= \boldsymbol{\theta}^T \langle T(\mathbf{z}, \mathbf{x}) \rangle_{q(\mathbf{z})} - \log Z(\boldsymbol{\theta}) + \text{const wrt } \boldsymbol{\theta} \end{aligned} \quad (30)$$

So, in the **E step** all we need to compute are the expected sufficient statistics under q .

Lemma 6.2 (Gradient of Log Partition Function).

$$\frac{\partial}{\partial \boldsymbol{\theta}} \log Z(\boldsymbol{\theta}) = \langle T(\xi) \rangle_{p(\xi|\boldsymbol{\theta})} \quad (31)$$

Proof.

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\theta}} \log Z(\boldsymbol{\theta}) &= \frac{1}{Z(\boldsymbol{\theta})} \frac{\partial}{\partial \boldsymbol{\theta}} Z(\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \frac{\partial}{\partial \boldsymbol{\theta}} \int f(\xi) \exp\{\boldsymbol{\theta}^T T(\xi)\} \\ &= \int \frac{1}{Z(\boldsymbol{\theta})} f(\xi) \exp\{\boldsymbol{\theta}^T T(\xi)\} \cdot T(\xi) = \langle T(\xi) \rangle_{p(\xi|\boldsymbol{\theta})} \end{aligned}$$

□

Corollary 6.3 (M Step for Exponential Families). *The M step (from Theorem 6.1, using Lemma 6.2) solves:*

$$\frac{\partial \mathcal{F}}{\partial \boldsymbol{\theta}} = \langle T(\mathbf{z}, \mathbf{x}) \rangle_{q(\mathbf{z})} - \langle T(\mathbf{z}, \mathbf{x}) \rangle_{p(\xi|\boldsymbol{\theta})} = 0 \quad (32)$$

6.1 EM for Exponential Family Mixtures

Example 6.4 (EM for Exponential Family Mixtures). For models with discrete latents (including mixtures, cf. Example 1.7), introduce an indicator vector $\mathbf{s}$ in place of discrete s :

$$s_i = m \quad \Leftrightarrow \quad \mathbf{s}_i = [0, 0, \dots, \underbrace{1}_{m\text{-th position}}, \dots, 0]$$

Collecting the M component distributions' natural parameters into a matrix $\boldsymbol{\Theta} = [\boldsymbol{\theta}_m]$:

$$\log P(\mathcal{X}, \mathcal{S}) = \sum_i [(\log \boldsymbol{\pi})^T \mathbf{s}_i + \mathbf{s}_i^T \boldsymbol{\Theta}^T T(\mathbf{x}_i) - \mathbf{s}_i^T \log \mathbf{Z}(\boldsymbol{\Theta})] + \text{const} \quad (33)$$

where $\log \mathbf{Z}(\boldsymbol{\Theta})$ collects the log-normalizers for all components into an M -element vector.

The expected sufficient statistics (E-step) are:

$$\begin{aligned} \sum_i \langle \mathbf{s}_i \rangle_q \quad & \text{(responsibilities } r_{im}) \\ \sum_i T(\mathbf{x}_i) \langle \mathbf{s}_i^T \rangle_q \quad & \text{(responsibility-weighted sufficient stats)} \end{aligned}$$

Maximization of the expected log-joint (M-step, cf. Corollary 6.3) gives:

$$\boldsymbol{\pi}^{(k+1)} \propto \sum_i \langle \mathbf{s}_i \rangle_q \quad (34)$$

$$\langle T(\mathbf{x}) | \boldsymbol{\theta}_m^{(k+1)} \rangle = \frac{\sum_i T(\mathbf{x}_i) \langle [\mathbf{s}_i]_m \rangle_q}{\sum_i \langle [\mathbf{s}_i]_m \rangle_q} \quad (35)$$

7 EM for MAP Estimation

Theorem 7.1 (EM for MAP). *Suppose we have a prior (with conjugate exponential family form as in Definition 1.1):*

$$p(\xi|\theta) = f(\xi) \exp\{\theta^T T(\xi)\}/Z(\theta), \quad p(\theta) = F(\nu, \tau) \exp\{\theta^T \tau\}/Z(\theta)^\nu \quad (36)$$

Augment the free energy (Proposition 2.6) by adding the log prior:

$$\begin{aligned} \mathcal{F}_{MAP}(q, \theta) &= \int q(\mathcal{Z}) \log p(\mathcal{Z}, \mathcal{X}, \theta) d\mathcal{Z} + H_q \leq \log P(\mathcal{X}|\theta) + \log P(\theta) \\ &= \int q(\mathcal{Z}) \left[\theta^T \left(\sum_i T(\xi_i) + \tau \right) - (N + \nu) \log Z(\theta) \right] d\mathcal{Z} + \text{const wrt } \theta \\ &= \theta^T (\langle T(\xi) \rangle_{q(\mathcal{Z})} + \tau) - (N + \nu) \log Z(\theta) + \text{const wrt } \theta \end{aligned} \quad (37)$$

The expected sufficient statistics in the E step are unchanged. After an E-step, the augmented free-energy equals the log-joint, so free-energy maxima are log-joint maxima (i.e., MAP values).

Remark 7.2 (Variational Bayes). Can we find posteriors? Only approximately – we’ll return to this later as “Variational Bayes”.

8 Time Series Models

8.1 Modeling Time Series

Definition 8.1 (Time Series). Consider a sequence of observations:

$$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_t$$

that are not independent.

Example 8.2 (Time Series Examples). • Sequence of images

- Stock prices
- Recorded speech or music, English sentences
- Kinematic variables in a robot
- Sensor readings from an industrial process
- Amino acids, DNA sequences

Remark 8.3 (Goal of Time Series Modeling). To build a joint probabilistic model of the data $p(\mathbf{x}_1, \dots, \mathbf{x}_t)$ to:

- Predict $p(\mathbf{x}_t | \mathbf{x}_1, \dots, \mathbf{x}_{t-1})$
- Detect abnormal/changed behavior (if $p(\mathbf{x}_t, \mathbf{x}_{t+1}, \dots | \mathbf{x}_1, \dots, \mathbf{x}_{t-1})$ is small)
- Recover underlying/latent/hidden causes linking entire sequence

8.2 Markov Models

Definition 8.4 (General Chain Rule). In general:

$$P(\mathbf{x}_1, \dots, \mathbf{x}_t) = P(\mathbf{x}_1)P(\mathbf{x}_2|\mathbf{x}_1)P(\mathbf{x}_3|\mathbf{x}_1, \mathbf{x}_2) \cdots P(\mathbf{x}_t|\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{t-1}) \quad (38)$$

Definition 8.5 (First-Order Markov Model). A first-order Markov model assumes:

$$P(\mathbf{x}_1, \dots, \mathbf{x}_t) = P(\mathbf{x}_1)P(\mathbf{x}_2|\mathbf{x}_1) \cdots P(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (39)$$

Definition 8.6 (Markov Property). The term *Markov* refers to a conditional independence relationship. In this case, the Markov property is that, given the present observation $(\mathbf{x}_t)$, the future $(\mathbf{x}_{t+1}, \dots)$ is independent of the past $(\mathbf{x}_1, \dots, \mathbf{x}_{t-1})$.

Definition 8.7 (Second-Order Markov Model). A second-order Markov model assumes:

$$P(\mathbf{x}_1, \dots, \mathbf{x}_t) = P(\mathbf{x}_1)P(\mathbf{x}_2|\mathbf{x}_1) \cdots P(\mathbf{x}_{t-1}|\mathbf{x}_{t-3}, \mathbf{x}_{t-2})P(\mathbf{x}_t|\mathbf{x}_{t-2}, \mathbf{x}_{t-1}) \quad (40)$$

where $\mathbf{x}_4$ is independent of $\mathbf{x}_1$ given $\mathbf{x}_2, \mathbf{x}_3$, but $\mathbf{x}_4$ may depend on $\mathbf{x}_2$ and $\mathbf{x}_3$.

8.3 Latent-Chain Models

Definition 8.8 (Latent-Chain Model). A latent-chain model captures temporal dependence through latent variables, with observations conditionally independent:

Joint probability factorizes:

$$P(\mathbf{z}_{1:T}, \mathbf{x}_{1:T}) = P(\mathbf{z}_1)P(\mathbf{x}_1|\mathbf{z}_1) \prod_{t=2}^T P(\mathbf{z}_t|\mathbf{z}_{t-1})P(\mathbf{x}_t|\mathbf{z}_t) \quad (41)$$

where $\mathbf{z}_t$ and $\mathbf{x}_t$ are both real-valued vectors, and $\square_{1:T} \equiv \square_1, \dots, \square_T$.

Remark 8.9 (Causal Structure). Temporal dependence is captured by latents (Definition 8.8), with observations conditionally independent. Applications include:

- **Speech recognition:** $\mathbf{z}$ = underlying phonemes or words, $\mathbf{x}$ = acoustic waveform
- **Vision:** $\mathbf{z}$ = object identities, poses, illumination; $\mathbf{x}$ = image pixel values
- **Industrial Monitoring:** $\mathbf{z}$ = current state of molten steel in caster; $\mathbf{x}$ = temperature and pressure sensor readings

8.4 Linear-Gaussian State-Space Models

Definition 8.10 (Linear-Gaussian State-Space Model). In a linear Gaussian SSM (a special case of Definition 8.8), all conditional distributions are linear and Gaussian:

Output equation: $\mathbf{x}_t = C\mathbf{z}_t + \mathbf{v}_t$

State dynamics equation: $\mathbf{z}_t = A\mathbf{z}_{t-1} + \mathbf{w}_t$

where $\mathbf{v}_t$ and $\mathbf{w}_t$ are uncorrelated zero-mean multivariate Gaussian noise vectors. Also assume $\mathbf{z}_1$ is multivariate Gaussian.

Remark 8.11 (Joint Gaussianity). The joint distribution over all variables $\mathbf{x}_{1:T}, \mathbf{z}_{1:T}$ is (one big) multivariate Gaussian.

Remark 8.12 (Alternative Names). These models (Definition 8.10) are also known as:

- Stochastic linear dynamical systems
- Kalman filter models

Remark 8.13 (Interpretation of SSMs). • Markov chain with linear dynamics $\mathbf{z}_t = A\mathbf{z}_{t-1}$

- Perturbed by Gaussian innovations noise (may describe stochasticity, unknown control, or model mismatch)
- Observations are a linear projection of the dynamical state, with additive iid Gaussian noise

Note. • Dynamical process ($\mathbf{z}_t$) may be higher dimensional than the observations ($\mathbf{x}_t$)

- Observations do not form a Markov chain – longer-scale dependence reflects/reveals latent dynamics

8.5 Hidden Markov Models

Definition 8.14 (Hidden Markov Model). Discrete hidden states $s_t \in \{1, \dots, K\}$; outputs $\mathbf{x}_t$ can be discrete or continuous.

Joint probability factorizes (cf. Definition 8.8):

$$P(s_{1:T}, \mathbf{x}_{1:T}) = P(s_1)P(\mathbf{x}_1|s_1) \prod_{t=2}^T P(s_t|s_{t-1})P(\mathbf{x}_t|s_t) \quad (42)$$

Generative process:

1. A first-order Markov chain (Definition 8.5) generates the hidden state sequence (path):
 - Initial state probabilities: $\pi_j = P(s_1 = j)$
 - Transition matrix: $\Phi_{ij} = P(s_{t+1} = j | s_t = i)$
2. A set of emission (output) distributions $A_j(\cdot)$ (one per state) converts state path to a sequence of observations $\mathbf{x}_t$:
$$A_j(\mathbf{x}) = P(\mathbf{x}_t = \mathbf{x} | s_t = j) \quad (\text{for continuous } \mathbf{x}_t)$$

$$A_{jk} = P(\mathbf{x}_t = k | s_t = j) \quad (\text{for discrete } \mathbf{x}_t)$$

Remark 8.15 (Two Interpretations of HMMs). 1. A Markov chain with stochastic measurements

2. A mixture model with states coupled across time

Remark 8.16 (Non-Markov Outputs). Even though the hidden state sequence is first-order Markov (Definition 8.6), the output process may not be Markov of any order (e.g., 1111121111311121111131...).

Remark 8.17 (Approximation Capacity). Discrete state, discrete output models can approximate any continuous dynamics and observation mapping even if nonlinear; however this is usually not practical.

Remark 8.18 (Relation to Automata). HMMs (Definition 8.14) are related to stochastic finite state machines/automata.

8.6 Comparison: HMMs and SSMs

Theorem 8.19 (HMMs vs SSMs). *Linear Gaussian state space models (Definition 8.10) are the continuous state analogue of hidden Markov models (Definition 8.14).*

Continuous vector state (SSMs):

- Very powerful representation
- For an HMM to communicate N bits of information about the past, it needs 2^N states!
- A real-valued state vector can store an arbitrary number of bits in principle

Linear-Gaussian output/dynamics (SSMs):

- Very weak
- The types of dynamics linear SSMs can capture is very limited
- HMMs can in principle represent arbitrary stochastic dynamics and output mappings

8.7 ML Learning for Chain Models with EM

Proposition 8.20 (Learning Structure for Chain Models). *The structure of learning and inference for both models (Definition 8.10 and Definition 8.14) is dictated by the factored structure (Definition 8.8):*

$$P(\mathbf{x}_1, \dots, \mathbf{x}_T, \mathbf{z}_1, \dots, \mathbf{z}_T) = P(\mathbf{z}_1) \prod_{t=2}^T P(\mathbf{z}_t | \mathbf{z}_{t-1}) \prod_{t=1}^T P(\mathbf{x}_t | \mathbf{z}_t) \quad (43)$$

Learning (M-step, cf. Definition 2.8):

$$\begin{aligned} & \arg \max \langle \log P(\mathbf{x}_1, \dots, \mathbf{x}_T, \mathbf{z}_1, \dots, \mathbf{z}_T) \rangle_{q(\mathbf{z}_1, \dots, \mathbf{z}_T)} \\ &= \arg \max \left[\langle \log P(\mathbf{z}_1) \rangle_{q(\mathbf{z}_1)} + \sum_{t=2}^T \langle \log P(\mathbf{z}_t | \mathbf{z}_{t-1}) \rangle_{q(\mathbf{z}_t, \mathbf{z}_{t-1})} + \sum_{t=1}^T \langle \log P(\mathbf{x}_t | \mathbf{z}_t) \rangle_{q(\mathbf{z}_t)} \right] \end{aligned} \quad (44)$$

So the expectations needed (in E-step) are derived from singleton and pairwise marginals.

9 Appendix: Proofs

9.1 KL Divergence is Non-Negative

Lemma 9.1 (KL Divergence Non-Negativity). *$KL[q(x)||p(x)] \geq 0$, with equality if and only if $\forall x : p(x) = q(x)$.*

Proof. Consider discrete distributions. The Kullback-Leibler divergence is:

$$KL[q||p] = \sum_i q_i \log \frac{q_i}{p_i} \quad (45)$$

To minimize with respect to distribution q , we need a Lagrange multiplier to enforce normalization:

$$E \equiv \text{KL}[q||p] + \lambda \left(1 - \sum_i q_i\right) = \sum_i q_i \log \frac{q_i}{p_i} + \lambda \left(1 - \sum_i q_i\right)$$

Find conditions for stationarity:

$$\begin{aligned} \frac{\partial E}{\partial q_i} &= \log q_i - \log p_i + 1 - \lambda = 0 \quad \Rightarrow \quad q_i = p_i \exp(\lambda - 1) \\ \frac{\partial E}{\partial \lambda} &= 1 - \sum_i q_i = 0 \quad \Rightarrow \quad \sum_i q_i = 1 \end{aligned}$$

Together these imply $q_i = p_i$.

Check sign of curvature (Hessian):

$$\frac{\partial^2 E}{\partial q_i \partial q_i} = \frac{1}{q_i} > 0, \quad \frac{\partial^2 E}{\partial q_i \partial q_j} = 0$$

So the unique stationary point $q_i = p_i$ is indeed a minimum. It is easily verified that at that minimum, $\text{KL}[q||p] = \text{KL}[p||p] = 0$.

A similar proof holds for continuous densities, using functional derivatives. $\square$

9.2 Fixed Points of EM are Stationary Points

Theorem 9.2 (Fixed Points are Stationary). *Fixed points of EM (Definition 2.1) are stationary points of $\ell(\boldsymbol{\theta})$.*

Proof. Let a fixed point of EM occur with parameter $\boldsymbol{\theta}^*$. Then:

$$\frac{\partial}{\partial \boldsymbol{\theta}} \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \Big|_{\boldsymbol{\theta}^*} = 0 \quad (46)$$

Now,

$$\begin{aligned} \ell(\boldsymbol{\theta}) &= \log P(\mathcal{X} | \boldsymbol{\theta}) = \langle \log P(\mathcal{X} | \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \\ &= \left\langle \log \frac{P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta})}{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta})} \right\rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \\ &= \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} - \langle \log P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \end{aligned}$$

Therefore,

$$\frac{d}{d\boldsymbol{\theta}} \ell(\boldsymbol{\theta}) = \frac{d}{d\boldsymbol{\theta}} \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} - \frac{d}{d\boldsymbol{\theta}} \langle \log P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \quad (47)$$

The second term is 0 at $\boldsymbol{\theta}^*$ if the derivative exists (minimum of $\text{KL}[\cdot||\cdot]$, by Lemma 9.1), and thus:

$$\frac{d}{d\boldsymbol{\theta}} \ell(\boldsymbol{\theta}) \Big|_{\boldsymbol{\theta}^*} = \frac{d}{d\boldsymbol{\theta}} \langle \log P(\mathcal{Z}, \mathcal{X} | \boldsymbol{\theta}) \rangle_{P(\mathcal{Z} | \mathcal{X}, \boldsymbol{\theta}^*)} \Big|_{\boldsymbol{\theta}^*} = 0$$

So, EM converges to a stationary point of $\ell(\boldsymbol{\theta})$. $\square$

9.3 Maxima in Free Energy Correspond to Maxima in Likelihood

Theorem 9.3 (Maxima Correspondence). *Maxima in $\mathcal{F}$ (Definition 2.5) correspond to maxima in ℓ .*

Proof. Let $\boldsymbol{\theta}^*$ be the parameter value at a local maximum of $\mathcal{F}$ (and thus at a fixed point).

Differentiating (47) with respect to $\boldsymbol{\theta}$ again:

$$\frac{d^2}{d\boldsymbol{\theta}^2}\ell(\boldsymbol{\theta}) = \frac{d^2}{d\boldsymbol{\theta}^2}\langle \log P(\mathcal{Z}, \mathcal{X}|\boldsymbol{\theta}) \rangle_{P(\mathcal{Z}|\mathcal{X}, \boldsymbol{\theta}^*)} - \frac{d^2}{d\boldsymbol{\theta}^2}\langle \log P(\mathcal{Z}|\mathcal{X}, \boldsymbol{\theta}) \rangle_{P(\mathcal{Z}|\mathcal{X}, \boldsymbol{\theta}^*)}$$

The first term on the right is negative (a maximum) and the second term is positive (a minimum, by Lemma 9.1). Thus the curvature of the likelihood is negative and $\boldsymbol{\theta}^*$ is a maximum of ℓ .

(Note: This holds as long as the derivatives exist. They sometimes don't, e.g., zero-noise ICA.) $\square$